

Data Governance in Data Engineering: Frameworks, Challenges, and Emerging Practices

Najma Shakirah binti Shahrulzaman

Faculty of Computing
Universiti Teknologi Malaysia
81310 UTM Johor Bahru, Malaysia
najmashakirah@graduate.utm.my

Nawwarah Auni binti Nazrudin

Faculty of Computing
Universiti Teknologi Malaysia
81310 UTM Johor Bahru, Malaysia
nawwarahauni@graduate.utm.my

Abstract—Data Governance plays an important role in the field of data engineering as it ensures that data is handled with policies and compliance in mind. Improper management of data without governance can cause business implications and issues related to data quality, security, and privacy. The complexity of data has made it more difficult to integrate data governance practices in organizations. DAMA-DMBOK is used as a globally recognized framework to structure data governance with other data management areas. The paper also highlights the advantages, limitations, and implementation challenges of data governance, as well as emerging trends. It is recommended to use the shift-left approach when developing data pipelines, automating quality checks, and defining data contracts between producers and consumers. Integration of data governance early in the data engineering process can result in resilient, scalable, and quality data assets.

Keywords: *Data Governance, Data Engineering, Data Quality, Data Security*

I. INTRODUCTION

Data governance in data engineering is a critical aspect of systems. It ensures that data is properly managed, secure, and reliable across different platforms and processes (Nai et al., 2025). In the big data age, organizations depend heavily on data to support decision-making, improve services, and gain competitive advantages. Because of this, managing data without proper governance can lead to serious issues such as poor data quality, security risks, and compliance problems (Wong & Mohamad, 2024).

In enterprise systems, data is generated from multiple sources. It includes business applications, customer interactions, and real-time systems. This makes data management more complex. Organizations must ensure that data is accurate, consistent, and accessible only to authorized users. Therefore, data governance plays an essential role by establishing policies, standards, and procedures that guide how data is handled throughout its lifecycle (Nai et al., 2025).

The main objective of this paper is to examine the role of data governance in data engineering and its importance in enterprise systems. It will explore key components such as data quality, data security, and governance policies. The scope of this paper focuses on governance practices within data engineering environments, including tools and frameworks

used in managing enterprise data. This paper also discusses real-world applications, benefits, challenges, and future trends of integrating data governance into modern data architectures.

II. BACKGROUND

Data governance has grown in complexity over the years due to the increasing number of stakeholders and activities associated with big data. It becomes difficult to access and manage data as a variety of data sources comes from many different administrative environments. Castro et al. (2021) highlights that data governance is necessary for managing actions that are involved with data including data engineering processes like data ingestion, transformation, storage and integration. The paper also makes a clear distinction between data governance that sets out the rules for data and data management that executes these rules when handling data. This signifies the data engineer's responsibility for designing and building pipelines that are governed and compliant.

To address this responsibility systematically, the field has developed structured frameworks that guide how governance should be organized and applied. The Data Management Body of Knowledge (DAMA-DMBOK) is a collection of processes that are regarded as the best practices within the data management areas, developed by DAMA International. As shown in Figure 1, it consists of 11 functional areas with data governance as the core. It provides a clear picture of governing data as a business asset and not a by-product of systems. This framework creates a structure that connects governance with architecture, quality, metadata, security and all functional data areas allowing governance to scale instead of breaking as systems get more complex (Mantha, 2020).

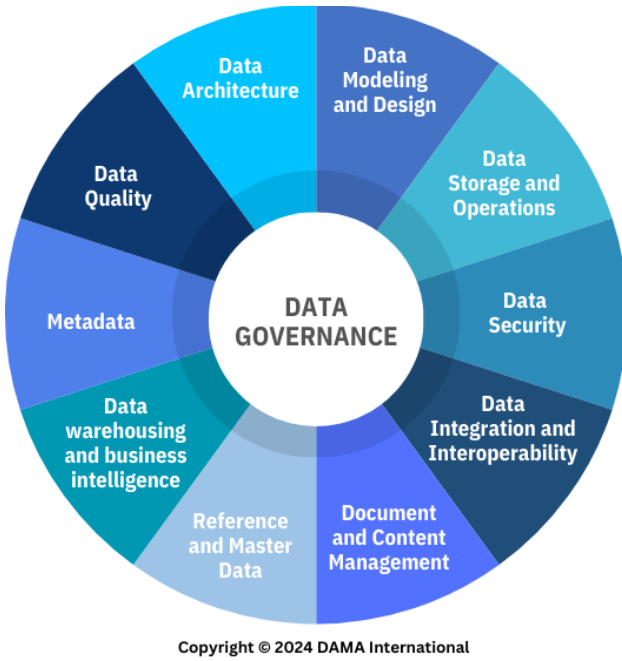


Figure 1: DAMA-DMBOK Wheel (DAMA International, 2017)

The practical impact of applying such governance structures within data pipelines is measurable across industries. For example, data governance ensures the data quality that is fed into the machine learning pipelines. In the banking industry, by applying automated data quality checks, lineage tracking, and standardization to banking data, it is shown to improve model accuracy by up to 15% and reduce false positives in fraud detection by 35% (Singamsetty, 2024). This demonstrates that governance embedded within the pipeline directly improves the reliability of the data products that engineers deliver.

However, a significant gap remains between how governance is traditionally practiced and the pace at which data engineering teams operate. Traditional governance methods are usually done where reviews only happen twice a year. Data engineers ship code and build pipelines almost daily or every week, and by the time the inspection is held, there are already problems and variation in policy implementation across tools. This mismatch encourages the adoption of data governance into data engineering workflows itself through practices such as policy-as-code and data contracts, treating compliance as a built-in engineering constraint rather than a periodic external review (Mantha, 2020).

III. ANALYSIS AND DISCUSSION

This section examines data governance through four lenses, how it functions within data engineering pipelines, real-world industry applications, its advantages and disadvantages, and the challenges and trends shaping its future. These dimensions provide a comprehensive understanding of how governance operates in practice and why its integration into data engineering workflows is increasingly critical.

A. How Data Governance Functions in Data Engineering

Data governance in data engineering is the set of policies, processes, standards, and technologies that ensure data is accurate, consistent, secure, and usable throughout an organisation's data pipelines. It operates as a framework that spans the full data lifecycle from ingestion and transformation to storage and consumption. Rules are enforced at every stage (Wong & Mohamad, 2024).

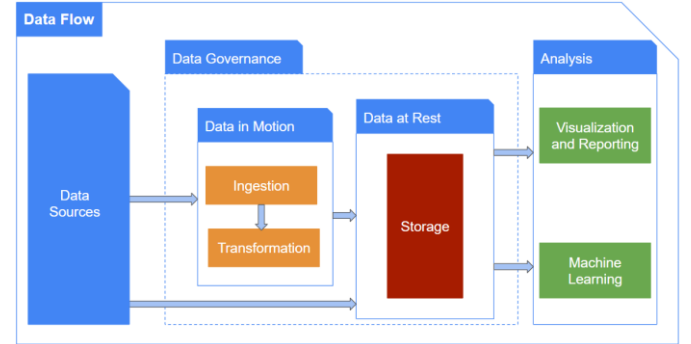


Figure 2: Data Governance Layer in a Typical Data Engineering Pipeline (Esbaï, 2023)

As illustrated in Figure 2, governance influences data ingestion by defining standards for data collection, validation, and compliance with regulations. As data moves through transformation, governance enforces data quality rules such as cleaning, normalization, and validation to ensure consistency and reliability. This helps prevent errors from propagating downstream and supports trustworthy analytics (Esbaï, 2023).

During the storage phase (data at rest), governance establishes policies for data security, access control, and privacy. It ensures that sensitive data is protected while remaining accessible to authorized users. Practices such as data cataloging and lineage tracking allow organizations to understand where data originates and how it evolves over time (Esbaï, 2023).

Finally, in the analysis layer, governance guarantees that only high-quality, well-documented data is used for reporting, visualization, and machine learning. This improves decision-making and builds trust in insights derived from the data (Esbaï, 2023). Data governance integrates people, processes, and technology to maintain data integrity and compliance across all stages of data engineering to produce reliable data-driven outcomes.

B. Case Examples and Real-World Applications

Data governance is applied across diverse industries. The following cases illustrate how real-world organisations have operationalised governance frameworks to achieve measurable outcomes:

1) Healthcare - Patient Data Integrity

Healthcare organisations in Malaysia face strict regulatory obligations under the Personal Data Protection Act (PDPA) 2010 while managing high-volume patient records. The Malaysian Health Data Warehouse (MyHDW) was introduced in 2017 by the Ministry of Health Malaysia to exchange patient medical records and expertise between public and private

hospitals and clinics, seeking to synchronise medical data from public hospitals, private hospitals and clinics, and the National Registration Department so that healthcare providers can make informed treatment decisions (Ghaleb et al., 2021).

Building on this, Malaysia has since become one of the first countries in the world to successfully implement the International Patient Summary (IPS) standard, with the long-term vision of a "1 Citizen, 1 Record" system where each Malaysian citizen has one consolidated health record accessible across all government and private health facilities (Peh, 2024). This directly reduces clinical errors and improves care coordination nationwide.

2) Retail - Customer Master Data Management

Retailers and e-commerce platforms in Malaysia employ data governance principles to unify fragmented customer records from online, in-store, and loyalty channels into a single consistent customer profile. Major retail operators such as AEON, as well as e-commerce platforms including Shopee and Lazada, manage large volumes of customer data across both physical and digital channels in Malaysia. Under the Personal Data Protection Act (PDPA) 2010, these organisations are legally required to govern how customer data is collected, stored, and used necessitating Master Data Management (MDM) practices to eliminate duplicates and maintain data accuracy (Chin, 2023). This enables accurate personalisation, improved inventory forecasting, and more reliable marketing analytics across all customer touchpoints.

C. Advantages and Disadvantages

Effective data governance delivers substantial organisational benefits, yet its implementation is not without challenges. Research by Lumenalta (2024) found that 98% of surveyed organisations reported that AI-driven data governance improved their decision-making processes, while 96% saw marked improvements in data quality after integrating AI into their governance frameworks. Table 1 provides a comparative overview of the key advantages and disadvantages.

Table 1 Advantages and Disadvantages of Data Governance (Lumenalta, 2024; Domo, 2024)

Advantages	Disadvantages
Improved decision-making through reliable high-quality data	High implementation cost and resource requirements
Enhanced data quality and consistency across the organization	Resistance to cultural change from staff and departments
Increased data security and regulatory compliance (GDPR, CCPA)	Managing data silos and fragmented legacy systems
Better data discovery and self-service analytics	Complexity in enforcing consistent policies enterprise-wide
Supports AI/ML initiatives with clean, trusted data	Risk of overly rigid governance stifling data

pipelines	innovation
-----------	------------

On the advantages side, Lumelnata (2024) notes that governance is critical for business intelligence. This enables data professionals to trust the data they rely upon for reporting and analytics. Domo (2024) further emphasises that governance tools support organisations in collecting, organising, and protecting data at scale which is a prerequisite for deploying AI and machine learning pipelines reliably. Conversely, the primary disadvantages relate to organisational friction are high setup costs, resistance from employees unaccustomed to governance policies, and the technical complexity of consolidating data spread across multiple legacy systems (Lumenalta, 2024).

D. Challenges and Future Trends

Data governance continues to evolve in response to technological change and regulatory pressure. Several key challenges and emerging trends are shaping the field heading into 2025 and beyond. The rapid rise of AI and unstructured data is becoming one of the challenges. Unlike traditional data governance which mostly addresses structured data, AI governance must handle a broader variety of data types including unstructured text, real-time streams, and synthetic data while accounting for how data is collected, labelled, processed, stored, and reused throughout the AI lifecycle (Lumenalta, 2024).

The field is evolving rapidly to meet these challenges head on. Machine learning and rules engines are increasingly automating quality checks and issue remediation. It reduces reliance on manual oversight. Metadata management and data lineage tools are giving organisations far greater visibility into how data flows and transforms across systems. Cloud-scale platforms are also maturing, offering built-in capabilities for access control, compliance monitoring, and lineage tracking out of the box (Achanta & Boina, 2023). These developments point toward a future where staying ahead in data governance means continuously adopting AI-driven tools and adapting to new trends, as scalability and real-time data processing become ever more critical (Lumenalta, 2024).

IV. RECOMMENDATIONS

This section discusses about recommendations that can be taken into consideration when integrating data governance in data engineering practices:

A. Integrating “Shift-left” approach into the data pipeline

Instead of doing the security checks after the code has been deployed and only catching the problems when things are too late, which can get really expensive and risky to fix (Mantha, 2020). Shift-left approach means to move the security and safety checks earlier in during the production of the system before any code gets deployed. This includes checking if the pipeline has the correct permissions, storage policies and scan transformation code if it has weak security. This avoids public buckets where anyone online can see the data storage and accidentally expose sensitive data like passwords and API keys that should have been encrypted.

B. Automated Quality Checks

Another strategy that should be implemented is using automated data quality checking. Instead of manual inspection, data quality rules should be embedded directly into the data transformation pipeline. As seen from the banking system use case, automated governance checks in the banking data pipeline successfully improved the machine learning model by up to 15% and reduced fraud detection false positives by 35%. Informatica Data Quality (IDQ) and Talend, are tools that can be used to test and validate data quality to ensure the data is accurate and reliable (Singamsetty, 2024).

C. Data contracts between producers and consumers

A data contract is a formal agreement that defines the expectations and responsibilities regarding data assets. Several components of data contracts include schema specification, semantic layer, quality expectations, operational parameters, governance metadata and overall implementation (Shashikumar, 2024). In data engineering, multiple teams produce and consume data and a schema change can break dashboards and machine learning models. Any changes that violate an existing contract will be blocked before it goes through the CI/CD pipeline.

Explicit agreements between producers and consumers will provide clarity and resilience to data ecosystems. There needs to be both changes in technical environment and cultural transformation when adapting to data contract principles. As organizations grow, data contracts will become an essential foundation to quality, trust and governance in data.

V. CONCLUSION

In conclusion, data governance plays a critical role in ensuring that organizational data is accurate, secure, consistent, and effectively managed across systems. The key takeaway from this study is that strong governance frameworks not only improve data quality and compliance with regulations such as PDPA but also enhance operational efficiency and decision-making. Another important insight is that organizations adopting mature data governance practices supported by clear policies, metadata management, and interoperability standards are better positioned to reduce data silos and improve trust in data assets.

A key lesson learned is that data governance is not solely a technical implementation but also a cultural and organizational responsibility requiring collaboration between stakeholders, data stewards, and leadership teams. Without governance alignment, even advanced data systems can produce unreliable or inconsistent outcomes.

Looking forward, future research could explore the integration of artificial intelligence and automation in data governance to improve real-time data quality monitoring and compliance enforcement.

ACKNOWLEDGMENT

First and foremost, we would like to take this opportunity to everyone that has helped us complete this technical paper. We wish to express my sincere appreciation to Dr. Aryati binti

Bakri, for encouragement, guidance, and critics throughout the process of this project.

Our sincere appreciation also extends to all our classmates and others who have helped on various occasions. Their views and tips are useful indeed. Without their continued support and interest, this project report would not have been the same as presented here.

REFERENCES

- [1] Achanta, A., & Boina, R. (2023). Data governance and quality management in data engineering. *International Journal of Computer Trends and Technology*, 71, 40–45. <https://doi.org/10.14445/22312803/IJCTT-V71I11P106>
- [2] Castro, A., Villagra, V. A., Garcia, P., Rivera, D., & Toledo, D. (2021). An Ontological-Based Model to data governance for big data. *IEEE Access*, 9, 109943–109959. <https://doi.org/10.1109/access.2021.3101938>
- [3] Chin J. (2023). PDPA 2010: Essential information for Malaysian companies. <https://www.qcloud.my/pdpa-2010-essential-information-for-malaysian-companies/>
- [4] Dama International. (2017). DAMA-DMBOK: Data Management Body of Knowledge (2nd Edition). Technics Publications, LLC, Denville, NJ, USA, pp. 71,
- [5] Domo. (2024). Data governance benefits. <https://www.domo.com/learn/article/data-governance-benefits>
- [6] Esbai, F. (2023). Data engineering 201: In-depth guide (Part 1). <https://www.firasesbai.com/articles/2023/03/11/data-engineering-201.html>
- [7] Ghaleb, E. A. A., Dominic, P. D. D., Fati, S. M., Muneer, A., & Ali, R. F. (2021). The Assessment of Big Data Adoption Readiness with a Technology–Organization–Environment Framework: A Perspective towards Healthcare Employees. *Sustainability*, 13(15), 8379. <https://doi.org/10.3390/su13158379>
- [8] Singamsetty, S. (2024). Efficacy of Data Governance a Cutting Edge Approach to Ensuring Data Quality in Machine Learning for Banking Industry. In 2024 2nd International Conference on Signal Processing, Communication, Power and Embedded System (SCOPES), 1–7. <https://doi.org/10.1109/scopes64467.2024.10991944>
- [9] Lumenalta. (2024). Benefits and challenges of data governance. <https://lumenalta.com/insights/benefits-challenges-of-data-governance>
- [10] Mantha, P. K. (2020). Integrating Data Governance and Security into Data Engineering Lifecycles: A Proactive Approach. *International Journal of AI BigData Computational and Management Studies*, 1(4). <https://doi.org/10.63282/3050-9416.ijaibdcms-v1i4p106>
- [11] Nai, S., Rifai, A., & Sadiq, A. (2025). Data Governance, Key Insights, Strategic Challenges, and Future Imperatives. In B. Elbaghazaoui, M. Amnai, & N. Gherabi (Eds.), *Data Governance, DevSecOps, and Advancements in Modern Software* (pp. 1-16). IGI Global Scientific Publishing. https://doi-org.ezproxy.utm.my/10.4018/979-8-3373-0365-9_ch001
- [12] Peh, C. (2024). Bridging healthcare's data divide: Malaysia pioneers global interoperability standards. GovInsider. <https://govinsider.asia/intl-en/article/bridging-healthcares-data-divide-malaysia-pioneers-global-interoperability-standards>
- [13] Shashikumar, N. (2024). An approach for redefining trust and accountability in modern data pipelines: Data contracts in the wild. *World Journal of Advanced Research and Reviews*, 24(2), 2888–2894. <https://doi.org/10.30574/wjarr.2024.24.2.3279>
- [14] Wong, H. M., & Mohamad, S. F. N. (2024). *The data governance: A comprehensive literature review from professional viewpoints*. Springer. <https://doi.org/10.13140/RG.2.2.16052.87683>